



Principles of Dimensional modeling

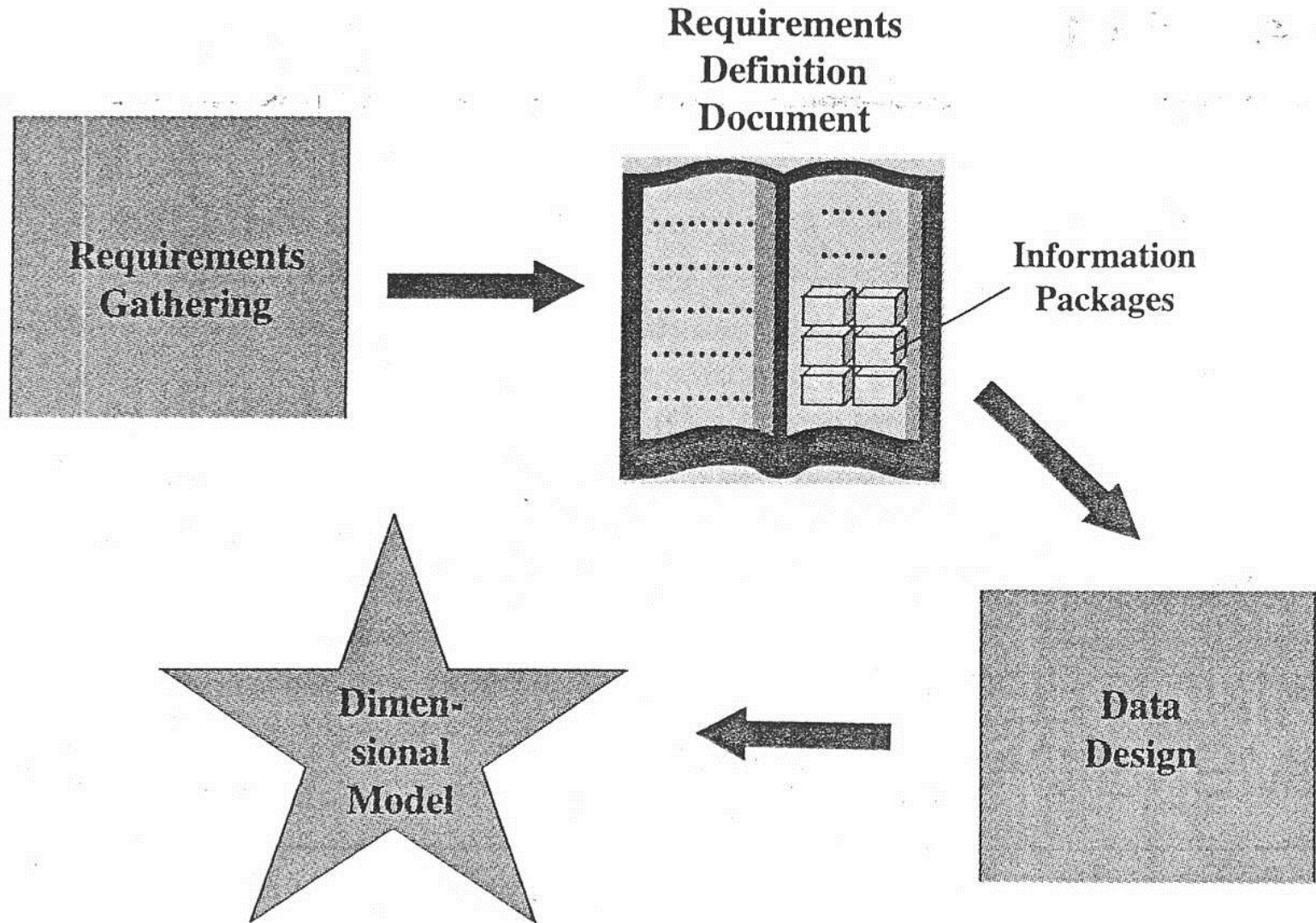




Today's lecture objective

- Requirement definition to design
- Star schema
 - Fact tables
 - Dimension tables
 - Surrogate key
 - Grain of fact table
- Advantages of star schema

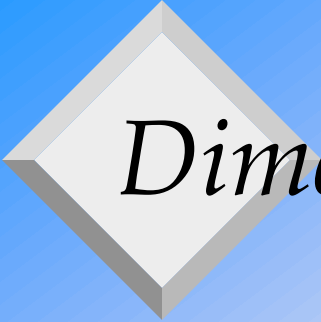
Requirement definition to design





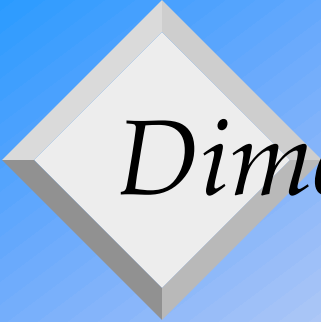
Design decision

- Choosing the process
 - Selecting the subjects from the information packages for the first set of logical structures to be designed
- Choosing the grain
 - Determining the level of detail for the data in the data structures
- Identifying and conforming dimensions
 - Making sure that each particular data element in every business dimension is conformed to one another
- Choosing the facts
 - Selecting the metrics or units of measures (eg, product sales unit, dollar sales, dollar revenue) to be included in first set
- Choosing the duration of the database
 - Determining how far back in time you should go for historical data.



Dimensional modeling basics

- It is logical design technique to structure the business dimensions and metrics that are analyzed along these techniques
- Gets its name from business dimensions
- Information Package is the foundation



Dimension tables

- It represents business dimensions
- Facts are used for the analysis
- The attributes in the dimension table acts as filters in our queries
- Each dimension table has an equal chance of a query
- Each dimension table has direct relationship with the fact table in the middle
- Each dimension table has one to many relationship with the fact table
- Such organization looks like a STAR

Dimensions

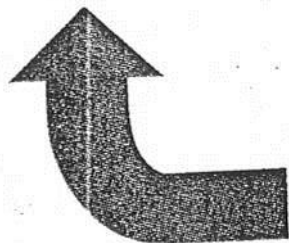
Automaker Sales

Fact Table

Actual Sale Price
MSRP Sale Price
Options Price
Full Price
Dealer Add-ons
Dealer Credits
Dealer Invoice
Down Payment
Proceeds
Finance

Time	Product	Payment Method	Customer Demo-graphics	Dealer	
Year	Model Name	Finance Type	Age	Dealer Name	
Quarter	Model Year	Term (Months)	Gender	City	
Month	Package Styling	Interest Rate	Income Range	State	
Date	Product Line	Agent	Marital Status	Single Brand Flag	
Day of Week	Product Category		Household Size	Date First Operation	
Day of Month	Exterior Color		Vehicles Owned		
Season	Interior Color		Home Value		
Holiday Flag	First Year		Own or Rent		

Facts: Actual Sale Price, MSRP Sale Price, Options Price, Full Price, Dealer Add-ons, Dealer Credits, Dealer Invoice, Down Payment, Proceeds, Finance



Dimension Tables

Product

Model Name
Model Year
Package Styling
Product Line
Product Category
Exterior Color
Interior Color
First Year

Time

Year
Quarter

Payment Method

Finance Type
Term

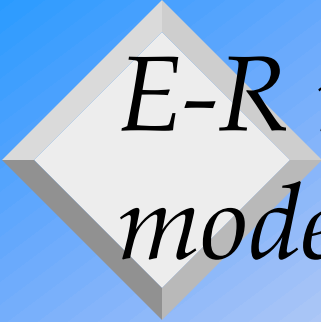
Customer Demographics

Age
Gender

Dealer

Dealer Name
City

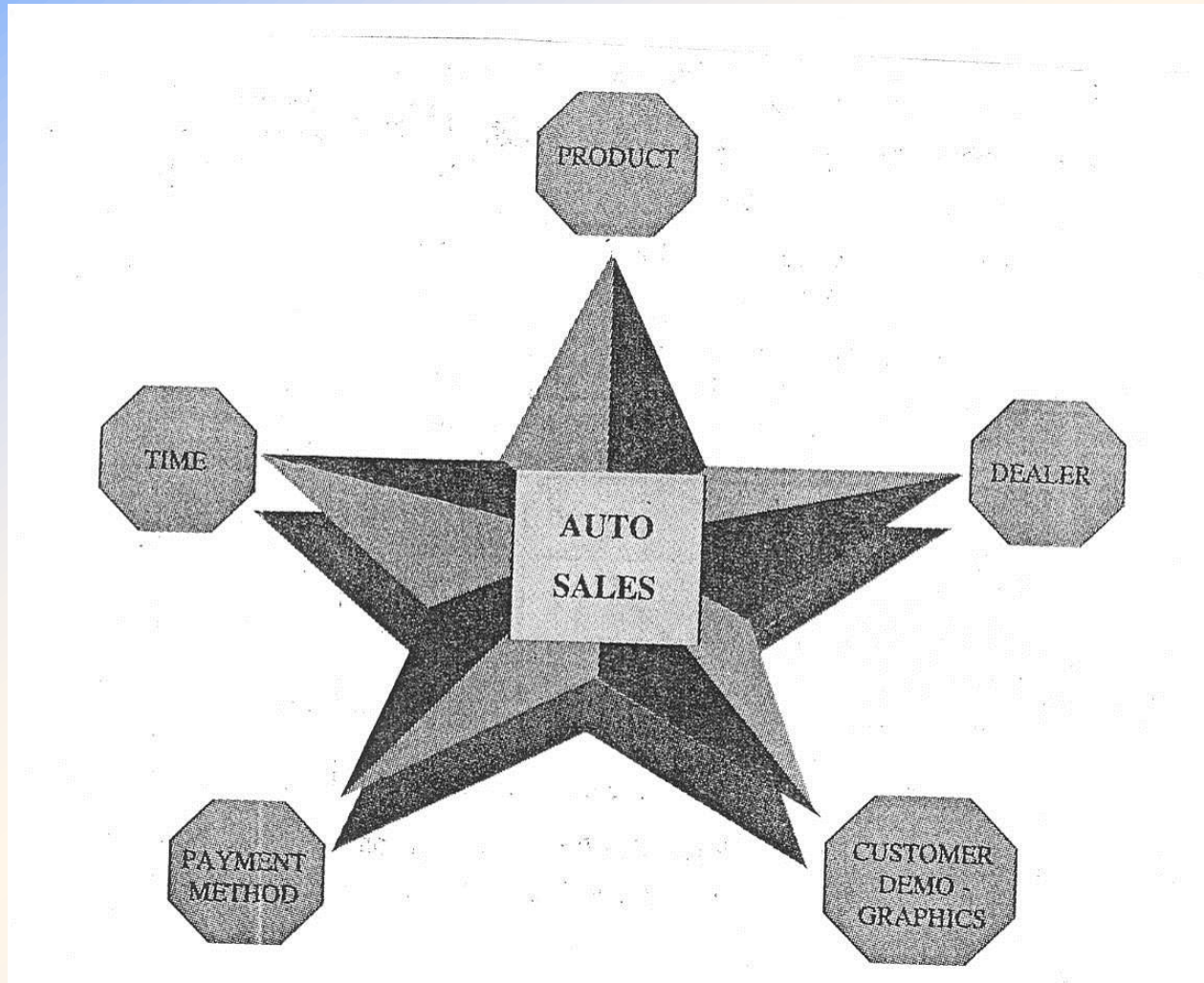
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E-R modeling and dimensional modeling

- E-R modeling
 - Removes the data redundancy
 - Ensures data consistency
 - Express microscopic relationships
 - For OLTP
- Dimensional modeling
 - Captures critical measures
 - Views along the dimensions
 - Intuitive to business users
 - For DWH

Star Schema





Star schema

➤ definition

- A simple database design in which dimensional data are separated from fact or event data (describing individual business transactions).
- Also known as dimension model
- Not suitable for ODS, OLTP, or EDWH
- Suitable for Ad-hoc queries
- Simplest star schema consists of one fact table surrounded by many dimension tables

➤ Fact table

- Contain factual or quantitative data about a business such as Units sold, order booked etc.
- PK of fact table is composite of FK



Star Schema Contd..

- Dimension table
 - Holds descriptive data about the subjects of the business
 - Dimension tables are demoralized



Inside a dimension table

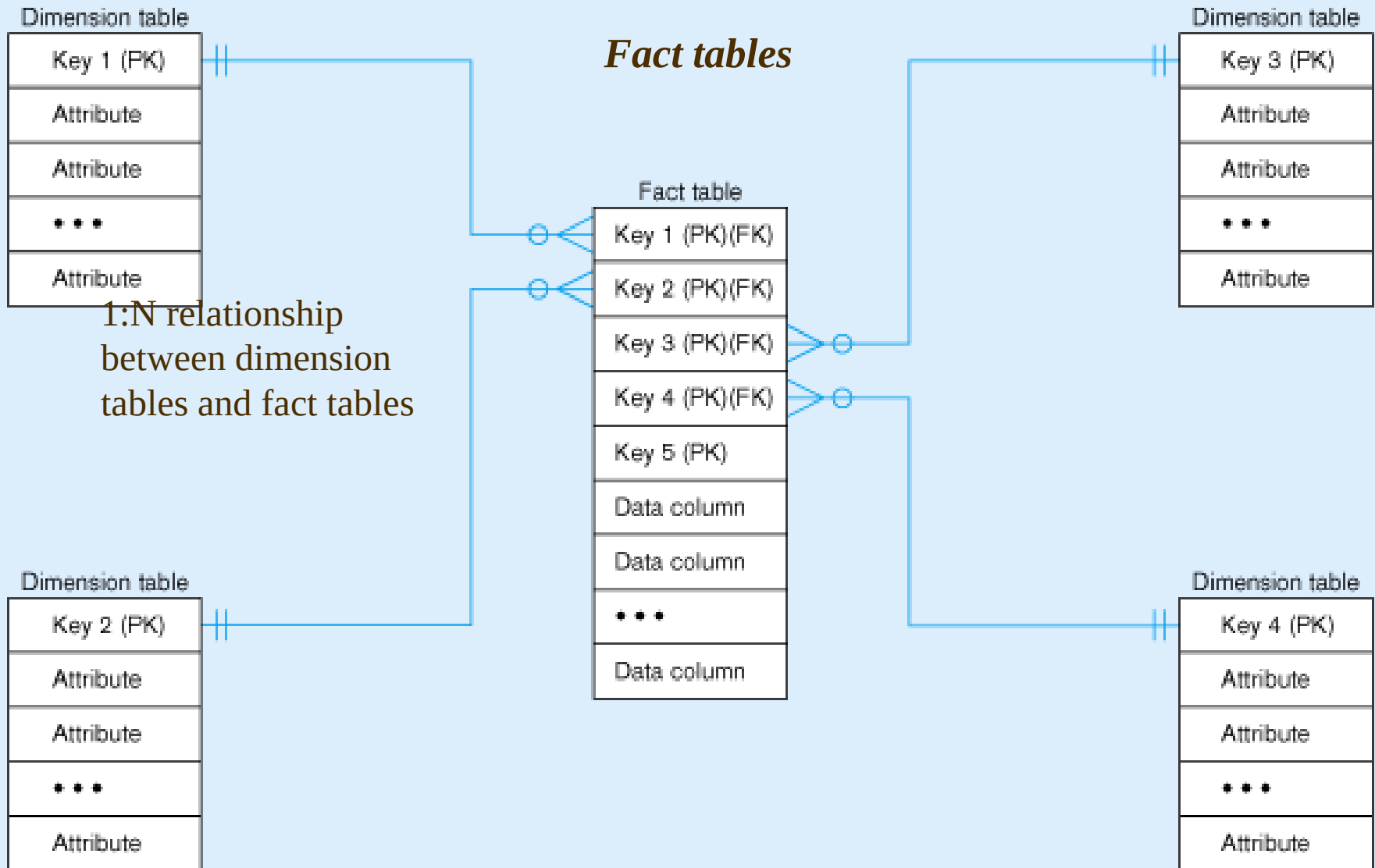
- Dimension table key
 - Primary key of dimension table key uniquely identifies each row
- Table is wide
- Fewer number of records
- Drilling down and rolling up
- Attributes not directly related



Inside the fact table

- Concatenated key
 - A row in the fact table relates to the combination of rows from all the dimensional tables
- Data grain
 - Data grain is the level of detail for the measurement metrics.
- Fully additive measures
 - Addition of values directly
- Semi additive measures
 - Addition of single values after calculation
 - Eg. Margin percentage
- Table deep not wide
- Sparse data

Components of a **star schema**



Star schema example

PRODUCT

<u>Product_Code</u>
Description
Color
Size

PERIOD

<u>Period_Code</u>
Year
Quarter
Month
Day

Fact table provides statistics for sales broken down by product, period and store dimensions

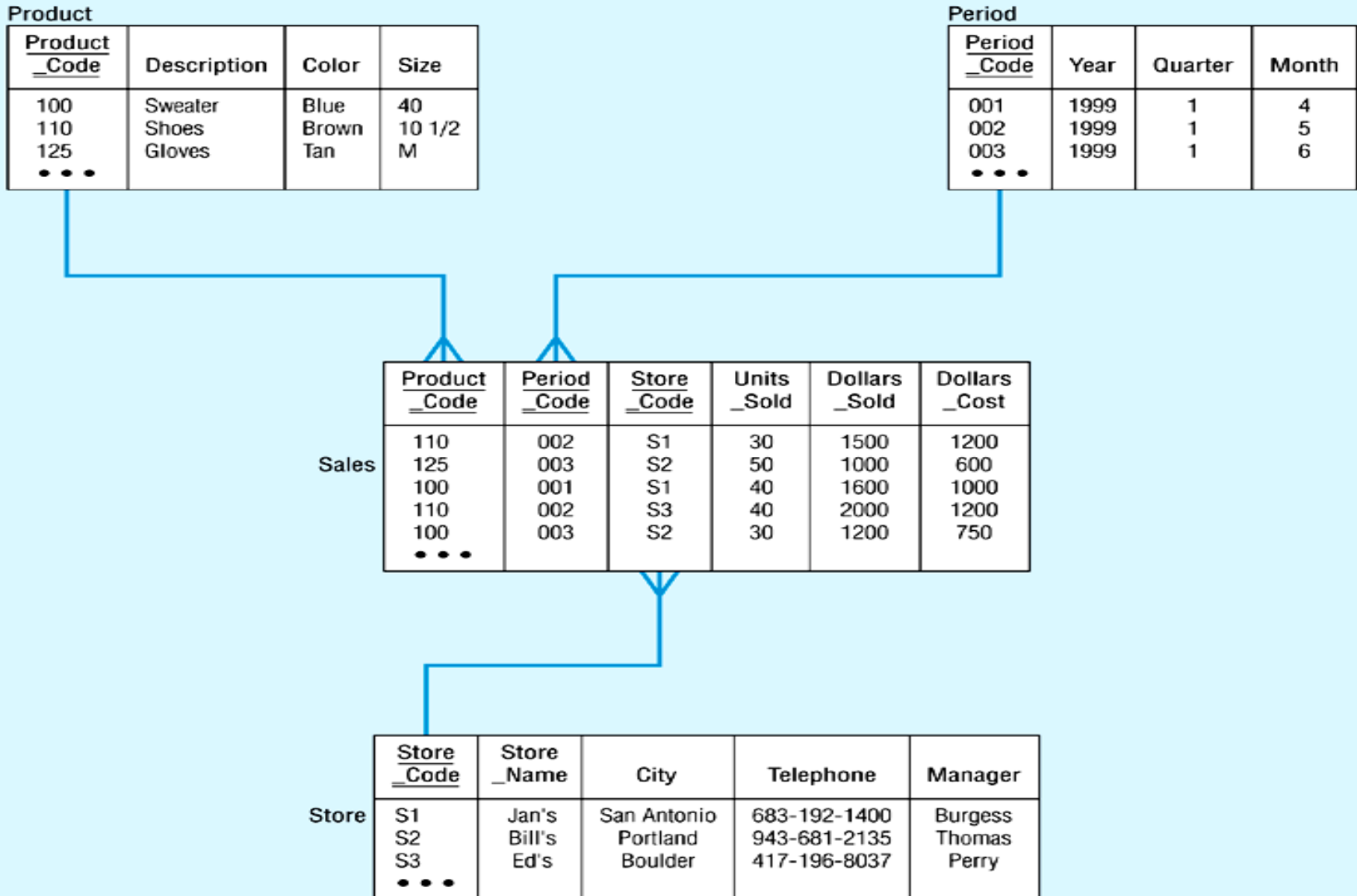
SALES

<u>Product_Code</u>
<u>Period_Code</u>
<u>Store_Code</u>
Units_Sold
Dollars_Sold
Dollars_Cost

STORE

<u>Store_Code</u>
Store_Name
City
Telephone
Manager

Star schema with sample data





Issues regarding star schema

- Surrogate key
 - Are simply system generated key
 - Don't use production keys as primary key for dimension table
 - The surrogate key will be mapped to the production key system



Grain of fact table



Advantages of star schema

- Easy for user to understand
 - Reflects exactly how user thinks
 - It is simple
 - Good because it is query centric structure
 - Ability to drill down and roll up.



Thank You Very Much